

# Multi-Agent Scientific Paper Review

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|                   |                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------|
| File              | Final_paper_kjns_210426 (1).pdf                                                                            |
| Words             | 12,275                                                                                                     |
| Sections Detected | preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra |
| Review Mode       | FAST                                                                                                       |
| Model             | qwen3:8b                                                                                                   |

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## Editorial Decision

MINOR REVISION

Weighted Score: 3.5 / 5.0  
Confidence: 82%

## Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

| Criterion                | Score | Rating    | Visual |
|--------------------------|-------|-----------|--------|
| Originality & Novelty    | 5.0   | Excellent | ■■■■■  |
| Manuscript Structure     | 2.5   | Weak      | ■■■■■  |
| Methodology & Equations  | 3.5   | Good      | ■■■■■  |
| Statistical Analysis     | 3.5   | Good      | ■■■■■  |
| Results & Evidence       | 4.0   | Good      | ■■■■■  |
| Discussion & Conclusions | 3.5   | Good      | ■■■■■  |
| Figures & Tables         | 4.0   | Good      | ■■■■■  |
| Scientific Writing       | 3.5   | Good      | ■■■■■  |
| References & Citations   | 2.5   | Weak      | ■■■■■  |
| Ethics & Transparency    | 1.5   | Very Weak | ■■■■■  |

# Detailed Agent Reviews

## TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 5.0 / 5.0 | Confidence: 95%

Strengths:

- [TITLE] effectively captures the novelty and technical focus of the study with specific terminology.
- [ABSTRACT] follows a structured format with clear background, objective, methods, results, and conclusions.
- [KEYWORDS] include a mix of specific and general terms, enhancing discoverability.

Minor Comments:

- [TITLE] could be slightly shortened for clarity without losing specificity.

Recommendations:

- [TITLE] consider shortening to improve readability while retaining technical specificity.

## StructureReviewer

Manuscript structure and logical coherence

Score: 2.5 / 5.0 | Confidence: 80%

Strengths:

- Clear section mapping
- Sections are generally well-organized

Weaknesses:

- Missing literature review section
- Lack of clear hypothesis or research question
- Section\_vi and section\_vii are duplicated content
- Conclusions are incomplete

Major Comments:

- The manuscript is missing a literature review section, which is critical for establishing the context and significance of the research.
- The hypothesis or research question is not explicitly declared, which weakens the clarity of the study's purpose.
- Section\_vi and section\_vii appear to be duplicates, which indicates a structural issue.
- The conclusions section is incomplete, which undermines the logical flow of the manuscript.

Minor Comments:

- The section mapping is not clearly defined in the manuscript.
- Some sections are not fully developed, such as the introduction and methodology.

Recommendations:

- Add a literature review section to provide context and background.
- Clearly state the hypothesis or research question in the introduction.
- Remove or revise the duplicated sections (section\_vi and section\_vii).
- Complete the conclusions section to ensure alignment with objectives and results.

## MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 3.5 / 5.0 | Confidence: 80%

Strengths:

- The integration of physical constraints into the neural mapping is a novel and important contribution.

- The use of multi-objective optimization for parameter identification is a well-justified approach.

**Weaknesses:**

- The problem statement is not clearly defined in the introduction.
- The methodology is not clearly separated from the experiments, making it difficult to follow the logical flow.
- The experimental setup and data preprocessing are not sufficiently detailed for reproducibility.
- The discussion and sections VI and VII are redundant and do not add new information.

**Major Comments:**

- The introduction lacks a clear problem statement and motivation for the proposed method.
- The methodology is not clearly separated from the experiments, making it difficult to follow the logical flow.

**Minor Comments:**

- The experimental setup and data preprocessing are not sufficiently detailed for reproducibility.
- The discussion and sections VI and VII are redundant and do not add new information.

**Recommendations:**

- Clarify the problem statement and motivation in the introduction.
- Separate the methodology from the experiments to improve the logical flow.
- Provide more details on the experimental setup, dataset, and preprocessing for reproducibility.
- Remove redundant sections (e.g., sections VI and VII) to improve clarity and conciseness.

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## StatisticsReviewer

*Statistical methods, tests, and reporting quality*

Score: **3.5 / 5.0** | Confidence: 80%

**Strengths:**

- Use of Kruskal-Wallis and post-hoc Wilcoxon/Holm corrections for non-parametric comparisons
- Bootstrap confidence intervals for robustness
- RMSE, MSE, and coefficient of variation for performance evaluation
- Pareto/hypervolume indicators for multi-objective optimization

**Weaknesses:**

- Lack of effect size reporting
- No verification of test assumptions
- Insufficient reproducibility details

**Major Comments:**

- The manuscript should provide evidence that the assumptions of the Kruskal-Wallis and Wilcoxon tests are met, or justify the use of non-parametric methods.
- Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to better interpret the magnitude of differences.
- Reproducibility details such as random seed values, number of runs, and variance estimates should be included to ensure transparency and replicability.

**Minor Comments:**

- Consider clarifying the interpretation of coefficient of variation and Pareto/hypervolume indicators to ensure they are correctly aligned with the study objectives.
- The manuscript should specify whether the 30 independent runs are sufficient for the statistical power required for the comparisons.

**Recommendations:**

- Include a power analysis to justify the sample size of 30 runs.
- Report effect sizes for all comparisons to complement p-values.
- Provide details on the random seed values and variance estimates for reproducibility.

- Verify the assumptions of the Kruskal-Wallis, and Wilcoxon tests (e.g., independence, distributional shape).
- Clarify the interpretation of coefficient of variation and hypervolume indicators in the context of the study.

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## FiguresTablesEquationsReviewer

*Figures, tables, equations — quality and coherence with text*

Score: 4.0 / 5.0 | Confidence: 80%

### Strengths:

- The manuscript includes a sufficient number of figures, tables, and equations for a Q1 journal.

### Weaknesses:

- The extracted text does not expose table bodies, so it is not possible to verify the completeness of the data or the presence of statistical significance indicators.

### Major Comments:

- The manuscript should provide the full content of tables and figures to ensure that they are self-explanatory and support the claims made in the text.

### Minor Comments:

- Ensure that all figures and tables are referenced in the text before they appear.

### Recommendations:

- Include the full content of tables and figures to verify data consistency and completeness.

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## ResultsReviewer

*Results validity and empirical support*

Score: 4.0 / 5.0 | Confidence: 80%

### Strengths:

- Comprehensive quantitative results including MSE, RMSE, CV, and hypervolume are reported
- Baseline comparisons with GA and PSO are included
- Pareto analysis and spread indicators are mentioned
- 30 independent runs are reported for statistical robustness

### Weaknesses:

- Negative results are not explicitly reported or acknowledged
- The results section is incomplete, with the text cut off before the full discussion of dynamic fidelity and parameter identification
- The connection between quantitative results and figures/tables is not clearly established

### Major Comments:

- The results section is incomplete and lacks full details on dynamic fidelity and parameter identification, which are critical for evaluating the framework's performance.

### Minor Comments:

- The connection between quantitative results and visual representations (figures/tables) is not clearly established.

### Recommendations:

- Complete the results section to fully address dynamic fidelity and parameter identification.

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## DiscussionConclusionsReviewer

*Discussion depth and conclusions validity*

Score: 3.5 / 5.0 | Confidence: 90%

### Strengths:

- Discussion provides a clear comparison of EANN with GA and PSO in terms of performance metrics and computational cost.

- Conclusions are directly tied to the experimental results and the research hypothesis.

**Weaknesses:**

- Discussion lacks a comprehensive comparison with prior work in the literature, focusing only on citation without actual comparison.
- Future work directions are not specified or justified in the conclusions section.

**Major Comments:**

- The discussion should include a more detailed comparison with existing literature to contextualize the findings and highlight the novelty of the EANN framework.

**Minor Comments:**

- The conclusion section could benefit from a more explicit statement on the contribution to the field.

**Recommendations:**

- Include a detailed comparison of the EANN framework with prior methods in the discussion to strengthen the contextualization of the findings.
- Specify future work directions in the conclusion section, such as potential applications or improvements to the EANN framework.

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## WritingReviewer

*Scientific writing quality, clarity, and style*

Score: **3.5 / 5.0** | Confidence: 80%

**Strengths:**

- Clear technical focus on physics-informed neural networks
- Well-structured methodology section

**Weaknesses:**

- Overuse of passive voice
- Lack of definition for EANN
- Vague quantifier 'significant'

**Major Comments:**

- The manuscript contains several instances of passive voice that could be rephrased for clarity and directness.
- The acronym EANN is used without definition, which may confuse readers unfamiliar with the term.

**Minor Comments:**

- The phrase 'significant' is used without numerical context, weakening the claim.
- Some sentences are overly long and could be split for better readability.

**Recommendations:**

- Define all acronyms on first use.
- Avoid passive voice where active voice is possible.
- Provide numerical context for vague quantifiers such as 'significant'.
- Break long sentences into shorter, more digestible ones.

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## ReferencesReviewer

*References quality, recency, and relevance*

Score: **2.5 / 5.0** | Confidence: 80%

**Strengths:**

- Recent references from 2021-2023
- Consistent citation style (IEEE format)

**Weaknesses:**

- Lack of seminal foundational works

- Missing key areas in literature review

**Major Comments:**

- The references section lacks seminal foundational works that are critical for understanding the theoretical basis of the research.
- The manuscript does not include key classic references on lumped mass-spring-damper models or structural health monitoring, which are central to the field.

**Minor Comments:**

- The references section includes a mix of conference and journal papers, which is acceptable, but more balance is needed.

**Recommendations:**

- Include seminal foundational papers from the 1990s to early 2000s that are foundational to structural modeling and identification.
- Add key references on structural health monitoring and observer-based state reconstruction to strengthen the literature review.

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## EthicsLimitationsReviewer

*Research ethics, limitations, and transparency*

Score: **1.5 / 5.0** | Confidence: 80%

**Strengths:**

- Limitations are explicitly stated in the discussion, indicating an awareness of the study's constraints.

**Weaknesses:**

- Limitations are not specific or detailed, which reduces the credibility of the study's self-assessment.
- The manuscript lacks a clear statement on reproducibility, including data and code availability.

**Major Comments:**

- The manuscript fails to report essential ethical and transparency information, which is critical for publication in Q1 journals.

**Minor Comments:**

- The limitations section could be more detailed and specific to strengthen the credibility of the study.

**Recommendations:**

- Include a statement on ethical approval for any human or animal studies.
- Provide a data availability statement to ensure transparency and reproducibility.
- Declare any potential conflicts of interest.
- Disclose funding sources.
- State the author contributions clearly.

## Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] The manuscript is missing a literature review section, which is critical for establishing the context and significance of the research.
- [2] The hypothesis or research question is not explicitly declared, which weakens the clarity of the study's purpose.
- [3] Section\_vi and section\_vii appear to be duplicates, which indicates a structural issue.
- [4] The conclusions section is incomplete, which undermines the logical flow of the manuscript.
- [5] The introduction lacks a clear problem statement and motivation for the proposed method.
- [6] The methodology is not clearly separated from the experiments, making it difficult to follow the logical flow.
- [7] The manuscript should provide evidence that the assumptions of the Kruskal-Wallis and Wilcoxon tests are met, or justify the use of non-parametric methods.
- [8] Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to better interpret the magnitude of differences.
- [9] Reproducibility details such as random seed values, number of runs, and variance estimates should be included to ensure transparency and replicability.
- [10] The manuscript should provide the full content of tables and figures to ensure that they are self-explanatory and support the claims made in the text.
- [11] The results section is incomplete and lacks full details on dynamic fidelity and parameter identification, which are critical for evaluating the framework's performance.
- [12] The discussion should include a more detailed comparison with existing literature to contextualize the findings and highlight the novelty of the EANN framework.

## Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [TITLE] could be slightly shortened for clarity without losing specificity.
- [2] The section mapping is not clearly defined in the manuscript.
- [3] Some sections are not fully developed, such as the introduction and methodology.
- [4] The experimental setup and data preprocessing are not sufficiently detailed for reproducibility.
- [5] The discussion and sections VI and VII are redundant and do not add new information.
- [6] Consider clarifying the interpretation of coefficient of variation and Pareto/hypervolume indicators to ensure they are correctly aligned with the study objectives.
- [7] The manuscript should specify whether the 30 independent runs are sufficient for the statistical power required for the comparisons.
- [8] Ensure that all figures and tables are referenced in the text before they appear.
- [9] The connection between quantitative results and visual representations (figures/tables) is not clearly established.
- [10] The conclusion section could benefit from a more explicit statement on the contribution to the field.
- [11] The phrase 'significant' is used without numerical context, weakening the claim.
- [12] Some sentences are overly long and could be split for better readability.

## Specific Improvement Recommendations

1. [TITLE] consider shortening to improve readability while retaining technical specificity.
2. Add a literature review section to provide context and background.
3. Clearly state the hypothesis or research question in the introduction.

4. Remove or revise the duplicated sections (section\_vi and section\_vii).
5. Complete the conclusions section to ensure alignment with objectives and results.
6. Clarify the problem statement and motivation in the introduction.
7. Separate the methodology from the experiments to improve the logical flow.
8. Provide more details on the experimental setup, dataset, and preprocessing for reproducibility.
9. Remove redundant sections (e.g., sections VI and VII) to improve clarity and conciseness.
10. Include a power analysis to justify the sample size of 30 runs.
11. Report effect sizes for all comparisons to complement p-values.
12. Provide details on the random seed values and variance estimates for reproducibility.

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